

Estimating root zone soil moisture using the SMAR model and regression method at a headwater catchment with complex terrain

Yongliang Qi ^{a,1}, Bihang Fan ^{a,1}, Yaling Zhang ^a, Yanjia Jiang ^a, Yuanyuan Huang ^a, Elizabeth W. Boyer ^b, Carlos R. Mello ^c, Li Guo ^a, Hongxia Li ^{a,*}

^a State Key Laboratory of Hydraulics and Mountain River Engineering, College of Water Resource and Hydropower, Sichuan University, Chengdu 610065, China

^b Department of Ecosystem Science and Management, Pennsylvania State University, University Park, PA 16802, USA

^c Water Resources Department, Universidade Federal de Lavras, CP 3037, Lavras, MG 37200-900, Brazil

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ABSTRACT

Obtaining accurate information regarding root zone soil moisture (RZSM) is a critical element of effective hydrological and agricultural management practices. Previous studies have relied on surface soil moisture (SSM) values, which are more easily measured, to estimate RZSM using the Soil Moisture Analytical Relationship (SMAR) model or regression method. However, the performance of these two types of methods in areas with complex topography still needs more exploration. Here, we assess the accuracy of these two types of methods in a forested mountainous catchment, using daily SSM measurements from 32 monitoring sites. The results show that both methods are capable of accurately estimating RZSM with a high NSE (>0.950) during the validation period. Additionally, they exhibit excellent model transferability at ungauged sites. Spatially, both methods perform better in drier areas than in wetter areas. Temporally, both methods are better in the wet–cold season than in the dry–warm season. Overall, both methods demonstrate comparable performance in the catchment, with NSE values of 0.986 and 0.951 during the validation period, respectively. The regression method is more suited to complex hydrological environments characterized by long-term soil moisture monitoring and nonlinear hydrological behaviors. Conversely, the SMAR model is better suited for flat areas and less spatial variability in microtopography. Moreover, the estimation of RZSM by both methods is influenced not only by soil moisture conditions but also by local factors including terrain topography, soil depth, and the degree of subsurface hydrological connectivity. This study adds to our understanding of RZSM estimation from SSM in complex terrain and will act as a reference for selecting appropriate methods of RZSM estimation. The results of this study underscore a discernible relationship between surface and deep soil moisture across varying spatial and temporal scales.

1. Introduction

Soil moisture (SM) is one of the most critical variables for agricultural and hydrological processes, which influences the exchange of water and energy between the land surface and the atmosphere (Kurc and Small, 2007; Li et al., 2021; Rigden et al., 2020). The root zone soil moisture (RZSM) content, which usually extends to the uppermost one meter of the soil profile, is especially important, as most hydrological and agricultural interests are related to the RZSM (Vereecken et al.,

2008). In addition, RZSM controls vegetation growth in water-limited ecosystems, such as shrublands, grasslands, and many temperate forests (Magagi and Kerr, 2001). Therefore, accurate quantification of RZSM is an important aspect of the attempts to improve climatic predictions, hydrological modeling, and agricultural planning (Kurc and Small, 2007; Pauwels et al., 2001) and adds to our understanding of ecohydrological and hydrological processes (Baldwin et al., 2017; Das and Mohanty, 2006; Li and Zhang, 2017; Li et al., 2009; Vereecken et al., 2008).

Abbreviations: DEM, Digital Elevation Model; NSE, Nash–Sutcliffe efficiency coefficient; PPF, Preferential flow frequency; SSM, Surface soil moisture; RZSM, Root zone soil moisture; SMAR, Soil Moisture Analytical Relationship.

* Corresponding author.

E-mail address: hongxiali@scu.edu.cn (H. Li).

¹ These authors contributed equally to this work.

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Direct SM measurements can be obtained by many approaches, such as traditional ground-based methods (Dobriyal et al., 2012; Wilson et al., 2003) and newly established remote sensing techniques (Dobriyal et al., 2012; Schmugge et al., 2002). Ground-based methods rely on direct contact between instruments and soil particles, which include the gravimetric method, thermostat-weight technique, neutron probe technology, and time-domain reflectometry method (Dobriyal et al., 2012). These methods often account for a limited number of samples from a specific location, which represents only a small area around the instruments (Bogena et al., 2007; Srivastava et al., 2013). Such techniques provide relatively precise data, but only point measurements are taken, which makes spatial interpretation difficult over a larger area (Western et al., 2002). Remote-sensing-based methods include the use of satellites, radar (microwaves), and active and passive sensors (Dobriyal et al., 2012; Vereecken et al., 2008). These methods can provide a spatial measurement of soil moisture over larger scales, e.g., catchment to regional scales. However, remote-sensing-based methods measure only the thin surface soil (i.e., depths of 0–0.05 m), so they cannot capture variations in RZSM, which extend to depths of 1 m. Therefore, accurate characterization of RZSM at the hillslope to catchment scales is still lacking.

Estimating methods are usually applied when direct RZSM measurements are challenging to acquire (Andrade et al., 2020). Many studies have shown that RZSM is correlated with environmental variables such as rainfall, air temperature, net radiation, and vegetation cover (Elshorbagy and Parasuraman, 2008). Thus, RZSM can be estimated based on environmental variables (Carranza et al., 2021). For example, Wang et al. (2007) developed a linear regression model to estimate RZSM from time series of Normalized Difference Vegetation Index (NDVI) obtained from semiarid and humid regions. Their results showed that the RZSM estimated from the deseasonalized NDVI data accounted for between 42 % and 71 % of the observed soil moisture variations. Carranza et al. (2021) developed two models (i.e., a Random Forest model and a process-based pore-flow model) over a small agricultural catchment using meteorological variables, including rainfall, temperature, relative humidity, and evapotranspiration as inputs. However, estimation of RZSM by environmental variables requires high quality data on environmental variables. In field monitoring, meteorological variables are often limited to a single monitoring point, and some environmental variables such as vegetation cover are discontinuously sampled, which will lead to poor model learning ability and unable to achieve fine prediction.

Previous studies have shown that RZSM can also be estimated using the surface soil moisture (SSM) values (depths of 0–0.10 m). This approach is more straightforward because SSM is relatively easy to measure and can be obtained from both in situ or satellite observations (Baldwin et al., 2019). The most common approach to estimating RZSM from SSM is the use of data-driven methods (Pezij et al., 2020), such as regression methods and artificial neural networks (ANNs) methods. For example, Ford et al. (2014) used cross-correlation analysis to quantify the association between SSM and RZSM. They found a strong relationship between near-surface (0.05–0.10 m) and root zone (0.25–0.60 m) soil moisture in the United States Great Plains. Pan et al. (2017) used the ANNs to retrieve RZSM at the continental scale using SSM data. They showed that the ANNs were effective in estimating RZSM at different depths, and a correlation coefficient above 0.7 was obtained at depths of 0.20 m in most cases.

Another method of RZSM estimation from SSM is to use physically based models, and the Soil Moisture Analytical Relationship (SMAR) model is the most widely used of these (Manfreda et al., 2014). Applications of the SMAR model have shown capability in estimating accurate RZSM values (e.g., Manfreda et al., 2014; Faridani et al., 2017). Baldwin et al. (2019) calibrated the SMAR model at 68 sites, and the average root mean square error (RMSE) for RZSM estimates from each satellite data product was $< 0.06 \text{ cm}^3 \text{ cm}^{-3}$. Farokhi et al. (2021) ran the SMAR model using three SSM input datasets with different spatial resolutions,

including ground-based SSM data and conventional and downscaled Advanced Microwave Scanning Radiometer 2 (AMSR2) SSM data. Their results showed that the SMAR model was capable of estimating RZSM, and its performance improved when using the downscaled SSM data. Baldwin et al. (2017) predicted RZSM using soil properties and satellite near-surface moisture data across the conterminous United States using SMAR-EnKF (Ensemble Kalman Filter), and their results showed that the predictions were in relatively close agreement with in situ observations. Because soil moisture was correlated with precipitation, evapotranspiration, soil depth, topography, and vegetation type, Liu et al. (2023) noted that SMAR model parameters were influenced by soil and climatic factors, and the applicability of the SMAR model varied at different depths in the rooting layer. Guo et al. (2023) suggested that the SMAR model relies heavily on evapotranspiration, which is influenced by a variety of factors including terrain, vegetation, and atmospheric conditions in addition to soil properties.

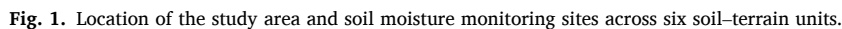
However, the performance of data-driven methods and physically based models to estimate RZSM is site-specific. The correlations between SSM and RZSM exhibit high variability under different environmental conditions (Mahmood and Hubbard, 2007). For example, plant water uptake is dependent on phenological stage and duration. Water use varies as a result of the various stages across the vegetation growing season. As a result, various land uses under the same climate have varied plant water consumption and, thus, RZSM distribution. Previous studies did not fully consider the influence of spatial heterogeneity of environmental factors on SSM, RZSM, and their relationship when estimating RZSM from SSM, and did not explore the difference of SSM-RZSM relationship across spatiotemporal scales. Therefore, we propose a hypothesis that even in a small catchment, due to the complex soil-topographic conditions, the relationship between SSM and RZSM has obvious spatial heterogeneity, and it is difficult to accurately estimate the RZSM and characterize its spatiotemporal pattern with a single model. Up to now, the adaptability of various methods (e.g., data-driven method and physically based model) under different spatio-temporal conditions is not clear. The estimation principles and application scenarios of the two models are different. The SMAR model considers physically significant hydrological processes, such as water infiltration, and does not include complex flow processes in the subsurface, such as subsurface lateral flow. The regression model is a data-driven model based on the characteristics of the monitoring data and partially reflects the impact of complex flow processes in the subsurface on the SSM-RZSM relationships. Therefore, it is essential to test the performance of these approaches in complex environments and determine the influence of various factors on RZSM estimation.

In the study, distributed long-term multi-layer soil moisture data at a headwater catchment are used to explore the spatial and temporal differences in the accuracy of SMAR and statistical models to derive RZSM from SSM, laying a foundation for more accurate application of these methods. This paper is concerned with the following questions: Which model, the SMAR or regression model, more accurately estimates RZSM in a headwater catchment with complex terrain? Further, this study investigates how the estimation accuracy of both models varies under diverse soil and topographic conditions, such as elevation, slope degree, slope curvature, soil texture, and soil depth. Additionally, it examines whether there are seasonal variations in the accuracy of RZSM estimations provided by the SMAR and regression models.

2. Study area and datasets

2.1. Shale hills catchment

Our study area (Fig. 1a) was located in the Shale Hills Critical Zone Observatory (CZO) catchment in central Pennsylvania, USA ($40^{\circ}39.87'N$, $77^{\circ}54.40'W$). The catchment area is 7.9 ha and the altitude ranges from 256 to 310 m above sea level. The valley is oriented in an E-W direction, dividing the catchment into the N- and S-facing aspects or



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The soil moisture data used from the study area were the daily observation data of soil moisture in the catchment from 19 May 2011 to 18 May 2014. For calibration, a two-year dataset spanning from 2011/5/19 to 2013/5/18 was employed, with the initial 15 days designated as the model warm-up period. Subsequently, a one-year dataset ranging from 2013/5/19 to 2014/5/18 was utilized for validation purposes.

2.2.2. Auxiliary datasets

Relevant information regarding soil texture was collected in conjunction with the soil moisture measurements. The soil properties associated with the geomorphological attributes and typical profiles at the 32 sites were observed, including slope curvature, slope degree, soil depth, elevation, frequency of preferential flow occurrence, and soil components (i.e., rock, sand, and clay). The value of each property for the entire soil profile was used for the site. Table 1 lists the datasets used in this study. We used pedological data, including soil texture, soil depth, and soil porosity (Lin, 2006; Takagi and Lin, 2012) and soil moisture measurements (Fan et al., 2020) to calibrate and validate the RZSM estimation models. We also used other datasets, such as preferential flow frequency (PFF), Digital Elevation Model (DEM), slope curvature, slope degree, and elevation, to investigate the possible influence on model performance.

3. Methods

3.1. Regression method

Given the possible close relationship between SSM and RZSM across soils and landscapes (Ford et al., 2014; Mahmood and Hubbard, 2007; Mahmood et al., 2012), we used SSM to estimate RZSM via regression method. Quadratic function (Eq. (1)) is used as it shows the best precision among the regression methods during the calibration and validation in this study:

$$\hat{y} = a \cdot x^2 + b \cdot x + c + \epsilon \quad (1)$$

where a , b , and $c(-)$ are coefficients derived by including x and y into the equation from routine in situ measurements of the calibration period; x is the SSM value of the site (horizon A); and y is the average RZSM value (horizons B and C). Using the in-situ measurements from the catchment, we developed regression equations for the soil moisture levels in the surface and root zones at each site. The RZSM estimations were calculated from the known a , b , c , and x values to obtain the RZSM values (y) at each site at that time.

Table 1
Datasets used in this study.

Data Attribute	Dataset	Time Range/ Temporal Resolution	Purpose	Reference
Hydropedology	Soil moisture	May 2011 to May 2014/ Daily	Calibrate/ validate estimation model	(Fan et al., 2020)
Hydropedology	PFF	May 2011 to May 2014/ Long-term average	Evaluate estimation model performance	(Liu and Lin, 2015)
Pedology	Soil texture, Soil depth, Soil porosity	One-time sampling	Establish SMAR model for Shale Hills	(Lin, 2006; Takagi and Lin, 2012)
Geology	DEM, Slope curvature, Slope degree, Elevation	One-time sampling	Evaluate estimation model performance	(Liu and Lin, 2015)

3.2. SMAR model

SMAR model is a physically based approach that is derived from a simplified soil water balance equation, providing a closed form of the relationship between the SSM and the RZSM with a limited number of physically consistent parameters (Manfreda et al., 2014). The SMAR model posits a two-layered soil structure comprising a surface layer, defined as the uppermost few centimeters of soil (at a minimum depth of 5 cm), and a root zone layer, which extends from the base of the surface layer down to the maximum rooting depth of the vegetation. The variables and parameters of the two layers are distinguished by footnotes "1" and "2". It simulates the infiltration of moisture from the surface layer into the root zone, as well as the soil water losses out from the root zone at a daily time-step (Manfreda et al., 2014). The only water mass exchange between the above horizons is infiltration. Other processes, including capillary rise and lateral flow, are assumed to be unimportant (Manfreda et al., 2014). Only when the surface soil moisture exceeds the field capacity will the water enter the root zone horizon. The water balance of the root zone horizon of the soil is controlled mainly by infiltration and water loss. The amount of water available to enter the root zone (y) ($-$) is equal to the water content exceeding a relative saturation value representing surface field capacity (s_{c1}) ($-$). It is assumed that the soil water loss linearly decreases from the maximum to 0 when the soil changes from full saturation to the wilting point. A time-discrete form of the SMAR model is given in Eq. (2).

$$s_2(t_j) = [s_{w2} + (s_2(t_{j-1}) - s_{w2})e^{-a(t_j - t_{j-1})}] + [(1 - s_{w2})by(t_j)(t_j - t_{j-1})] \quad (2)$$

$$y(t_j) = \begin{cases} s_1(t_j) - s_{c1}, & s_1(t_j) \geq s_{c1} \\ 0, & s_1(t_j) < s_{c1} \end{cases} \quad (3)$$

Here, $s_2(t)$ ($-$) is the soil moisture at the root zone relative to the saturation moisture at time t_j , s_{w2} ($-$) is the soil moisture in the wilting point of the root zone relative to the saturation moisture (the range is between 0–1), $y(t)$ ($-$) is the infiltration part of the root zone, $s_1(t)$ ($-$) is the soil moisture at surface horizon relative to the saturation moisture at time t_j , s_{c1} ($-$) is the value of relative saturation at field capacity of the surface horizon of soil (changing between 0–1), a ($1/T$) represents the water loss coefficient, and b ($-$) represents the diffusion coefficient that controls excess water entering the root zone, j is the Julian day. The coefficients a ($1/T$) and b ($-$) can be estimated using Eq. (4):

$$a = \frac{V_2}{(1 - s_{w2})n_2Z_{r2}} \quad b = \frac{n_1Z_{r1}}{(1 - s_{w2})n_2Z_{r2}} \quad (4)$$

where V_2 (LT^{-1}) is the water loss coefficient of the root zone, n_1 ($-$) is the soil porosity of the surface horizon, Z_{r1} (L) is the soil thickness of the surface horizon, n_2 ($-$) is the soil porosity of the root zone, and Z_{r2} (L) is the soil thickness of the root zone.

In this study, the daily SSM (mainly distributed in 0–10 cm depth) was measured by in situ sensors in the surface layer (horizon A), and the daily RZSM (mainly distributed in 10–100 cm depth) was equivalent to the average measured SM by sensors at horizons B and C. The time series of the relative SSM of the site served as the input data for the SMAR model and outputs estimates of the RZSM. Fig. 2 shows the flowchart of the SMAR model structure. The genetic algorithm toolbox within MATLAB was used to optimize the model parameters (Dong, 2008). The parameters a , b , s_{w2} and s_{c1} were calibrated (Baldwin et al., 2019) and the minimum RMSE (Eq. (7)) between the simulated and observed relative RZSM at the site was used as the optimization objective.

3.3. Regionalization method

The performance of the two models was further evaluated by testing the potential for RZSM estimation at ungauged/unmonitored sites/

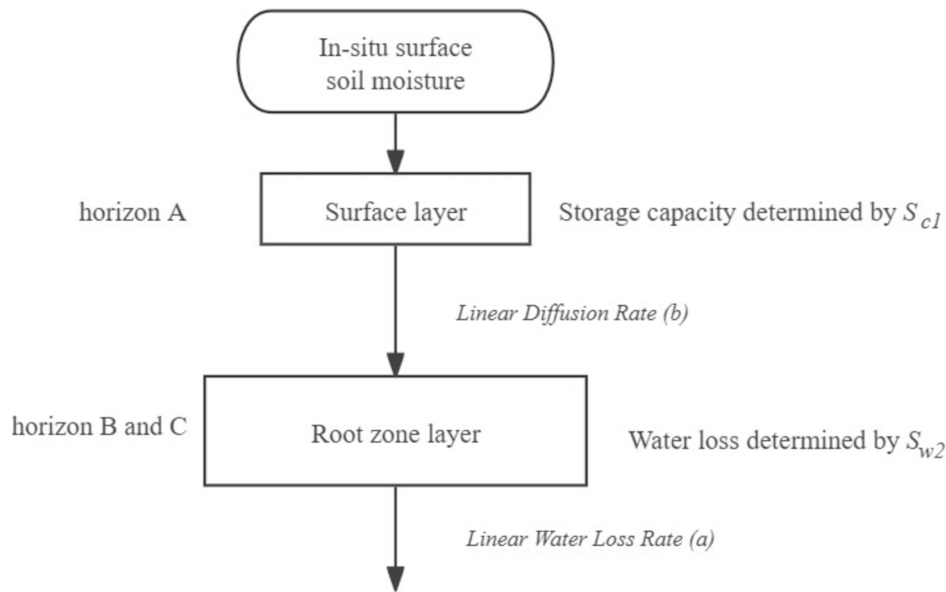


Fig. 2. Flowchart of the soil moisture analysis relationship (SMAR) model, with arrows indicating the sequence of calculations during each modeling process.

regions by regionalization approach (Li et al., 2009; Parajka et al., 2005). That is, transferring calibrated model parameters from gauged (donor) to ungauged (target) sites for estimation (Merz and Blöschl, 2004; Oudin et al., 2008). Our regionalization was carried out in the six soil–terrain units where the sites were grouped into one of six soil–terrain units (as described in 2.1). Each of the sites in the group was left out and regarded as an ungauged site, and the parameter sets obtained from the donor (other sites in the group) sites are used to simulate RZSM in the targeted ungauged site. For the evaluation of the robustness of regionalization, every site in the group was considered a donor.

3.4. Model accuracy evaluation index

The ability of two types of methods to simulate RZSM was assessed using the Nash–Sutcliffe efficiency coefficient (NSE), the determination coefficient (R^2), and RMSE, as shown in Eqs. (5)–(7).

$$NSE = 1 - \frac{\sum_{i=1}^n (x_o^i - x_s^i)^2}{\sum_{i=1}^n (x_o^i - \bar{x}_o)^2} \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (x_o^i - x_s^i)^2}{\sum_{i=1}^n (x_s^i - \bar{x}_s)^2} \quad (6)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (x_o^i - x_s^i)^2}{n}} \quad (7)$$

Here, x_o is the observed value, x_s is the simulated value, $\bar{x}_o$ is the average value from the observations, and $\bar{x}_s$ is the average value from the simulation. NSE and R^2 represent the correlation between the simulated and observed RZSM. A high value of NSE and R^2 indicates good performance in estimating RZSM. The RMSE represents the deviation between the observed value and the estimated value. When the estimated value is the same as the observed value, the RMSE equals 0; the greater the difference, the higher the RMSE. A high NSE , high R^2 , and low RMSE together indicate that the estimate is as accurate as possible.

Due to the SMAR simulation mechanism, error indicators at the soil unit and site scales are based on the relative soil moisture content rather than the raw volumetric water content. Therefore, our accuracy evaluation of the regression method was also converted from the volumetric

soil moisture content to the relative soil moisture content (i.e., the conversion of the two datasets using soil porosity data). This ensured that both methods had a similar range of means and values for the estimation performance assessment. We chose different datasets to assess the models' performance at various scales. At the catchment scale, we make use of soil volumetric water content data. Data on relative SM are used for analysis at the soil–terrain unit and site scales. Further, we used the volumetric soil moisture content for the performance evaluation at the whole catchment scale.

3.5. Statistics

We used nonparametric statistical analysis to compare the differences between groups (with more or less accurate RZSM estimation) within the datasets, as they showed heterogeneity or non-normality. We used the Mann–Whitney test for the non-parameter statistical analysis when only two groups were involved and the Kruskal–Wallis test when more than two groups were involved. This analysis was carried out using the SPSS software (Okech, 2009). In this study, the Mann–Whitney test was used to determine whether there were any significant variations in the soil characterization factors between the two groups (low accuracy and high accuracy of RZSM estimation) of study sites. The differences were statistically significant when $p < 0.05$.

4. Results

4.1. Spatial assessment of estimation ability of two types of methods

The assessment ability of two types of methods was evaluated across the whole catchment, in the six soil–terrain units, and at the 32 sites of soil moisture monitoring. Table S2 describes the values of the SMAR model parameters for the 32 sites. Fig. 3 compares the mean values of the observed SSM and RZSM values during the validation period (2013/5/19 to 2014/5/18) with the estimated RZSM values derived from both methods. These estimations, applied to the entire catchment area, were interpolated using ordinary kriging. Compared with the observed RZSM, the observed SSM is wetter, with no evident spatial pattern. In contrast, the observed RZSM shows evident spatial variations, with wetter areas along the valley and drier areas along the ridgetop (Fig. 3a, b). Overall, both methods provide reasonable estimations of RZSM (Fig. 3b, c, d). However, compared with the observations (Fig. 3b), the RZSM

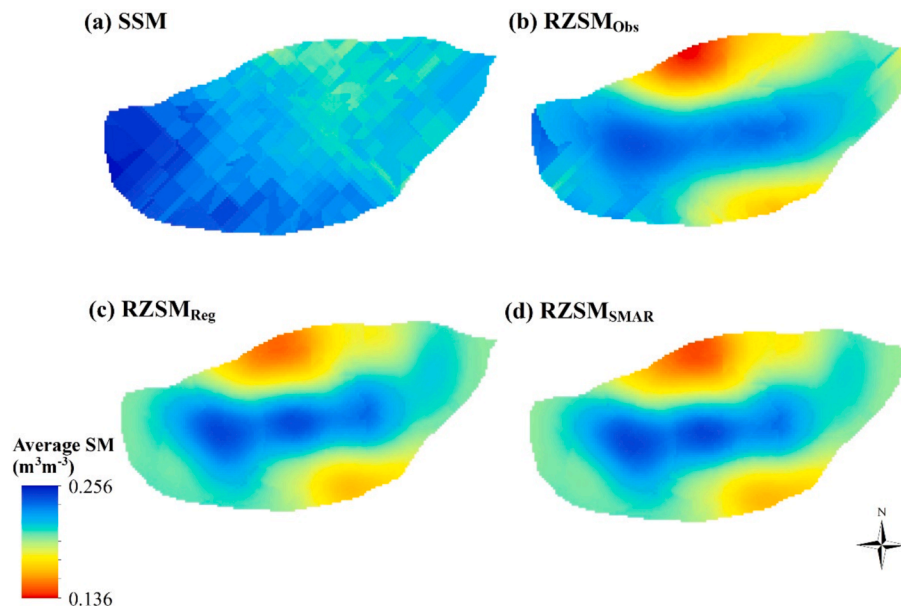


Fig. 3. Distribution of the mean observed (a) SSM and (b) RZSM, (c) estimated RZSM by regression, and (d) SMAR methods in the Shale Hills catchment during the validation period.

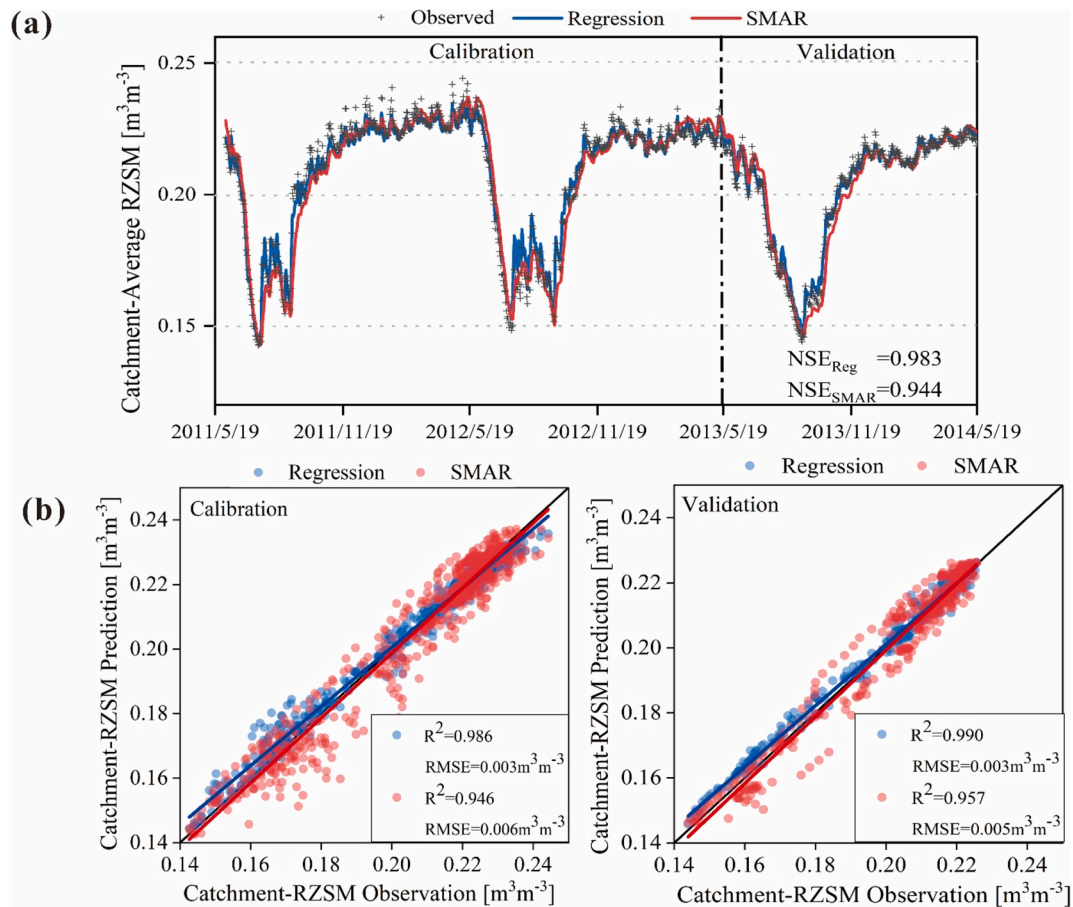


Fig. 4. (a) Time series of daily catchment-average RZSM estimated by the regression (blue curve) and SMAR (red curve) methods against field observations (crosses) in the Shale Hills catchment from 2011 to 2014. (b) Scatter plots and accuracy metrics of observed and estimated values based on regression (blue dots) and SMAR (red dots) methods for the calibration (2011–2013) and validation periods (2013–2014). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

estimations by both methods show a greater spatial contrast, with wetter values in the valley (Fig. 3c, d).

When comparing the observations, the ridge area is mostly where the differences between the two models are focused. Additionally, the simulated values of the two approaches were associated with the observed values of average root zone soil moisture at 32 sites to quantitatively explain the comparison results. Consequently, 0.976 and 0.972 were the correlation coefficients between the observed values and the regression and SMAR estimated values.

Fig. 4 compares the catchment-average observed and estimated RZSM values from 2011 to 2014. The results show that both methods provide good estimations (Fig. 4a). The NSE values of the regression method and the SMAR are 0.986 and 0.951 in the validation period, respectively, both with high accuracy. Fig. 4b shows the 1:1 correspondence between the average volumetric soil water content determined from the in situ measurements and two types of methods. Results from two types of methods generally have high R^2 (>0.940) and low RMSE ($<0.006 \text{ m}^3/\text{m}^3$) values, demonstrating that the estimations are similar to the observations. Both methods accurately estimate the long-term average RZSM, with the regression method slightly outperforming the estimation capabilities of the SMAR model.

The ability of two types of methods to estimate RZSM was also evaluated in the six soil-terrain units (Fig. 5). Compared with the measured relative soil moisture value, the results show that the R^2 values for the six different units from the regression method range from 0.872 to 0.967, and the RMSE ranges from 0.007 to 0.022. In comparison, the R^2 of the SMAR method varies from 0.805 to 0.941, and RMSE ranges from 0.014 to 0.025. These results demonstrate that both the regression and SMAR approaches perform well, and the regression method is a little accurate (Fig. 5).

Two types of methods of simulating RZSM(–) were also assessed at

the site scale, and the RMSEs from all sites are ranked in Fig. 6a. We set an RMSE(–) of 0.05 as the acceptable model performance criteria, and approximately 80 % of the sites met this RMSE benchmark across the catchment. Two types of methods show similar estimation accuracy at most sites, but with slightly better performance than the regression method. The SMAR model show favorable performance on flat shaded aspect (Fig. 6b). The calibration and validation accuracies of the SMAR and regression models are supplemented in Table S3. Fig. 6b shows the sites that did (green dots) and did not (yellow dots) meet the RMSE benchmark. The regional RZSM estimations from the regression and SMAR methods show similar spatial trends. The majority of sites with $\text{RMSE} < 0.05$ are grouped in the drier areas of the region, whereas sites with $\text{RMSE} > 0.05$ are located mainly in the valley where wetness is higher.

4.2. Temporal assessment of the estimation ability of two types of methods

We also investigated the RZSM estimations from two types of methods during different seasons. Fig. 7a displays scatter plots for the catchment's dry–warm (red dots) and wet–cold (green dots) seasons during the validation period. Both the wet–cold and dry–warm seasons show a concentration around the 1:1 line, indicating general agreement between the observed and estimated values and, therefore, good correspondence between the two datasets. The RMSE values for the dry–warm and wet–cold seasons are 0.004 and $0.002 \text{ m}^3/\text{m}^3$, respectively, for the regression method and 0.006 and $0.004 \text{ m}^3/\text{m}^3$, respectively, for the SMAR method. This demonstrates that the regression method performs slightly better in capturing the moisture dynamics than the SMAR approach. Both methods are more effective for estimating RZSM during the wet–cold season. During the dry–warm season, the estimations of SMAR are more dispersed, with some underestimated RZSM

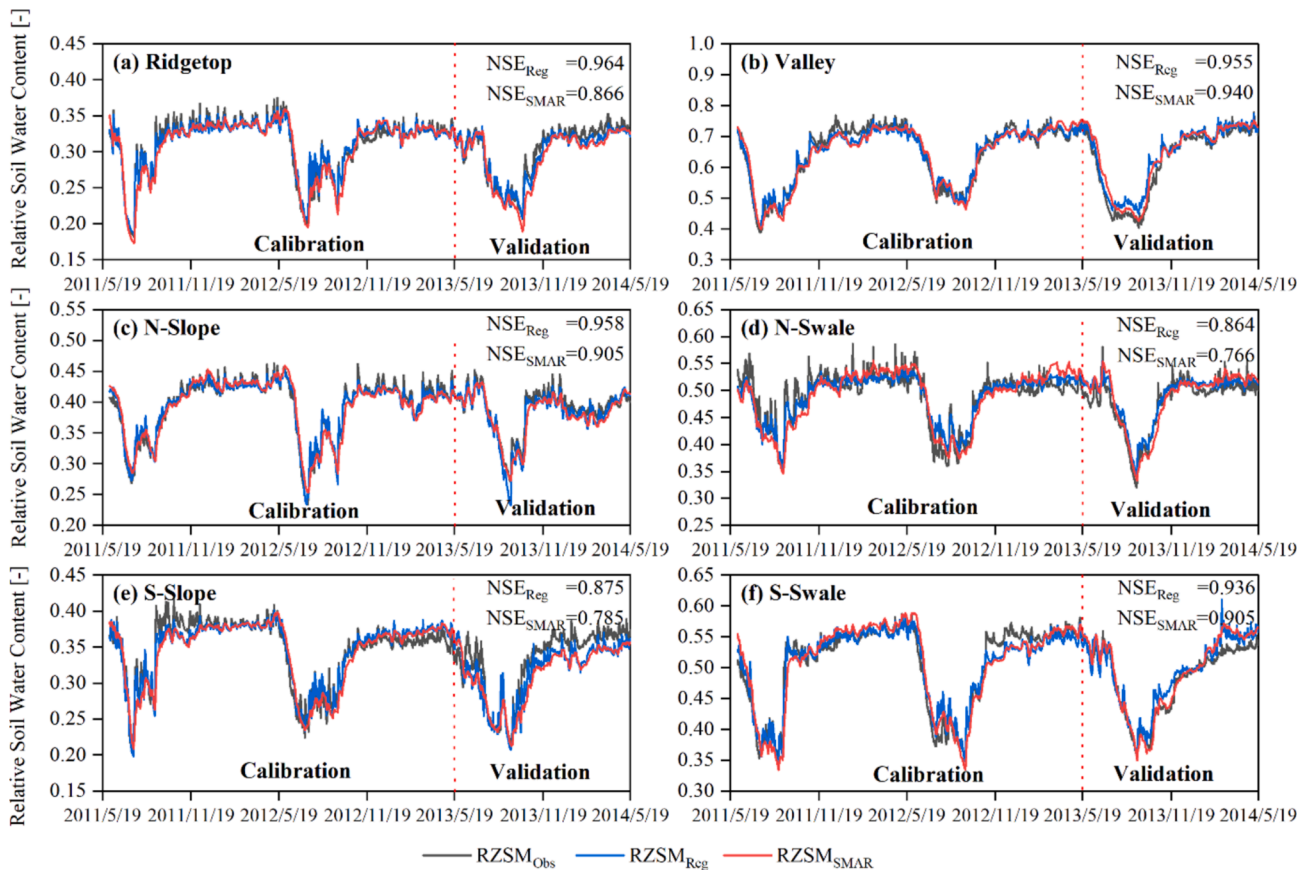


Fig. 5. Time series of daily catchment-average RZSM estimated by the regression (blue curve) and SMAR (red curve) methods against field observations (black curve) for the six soil-terrain units. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

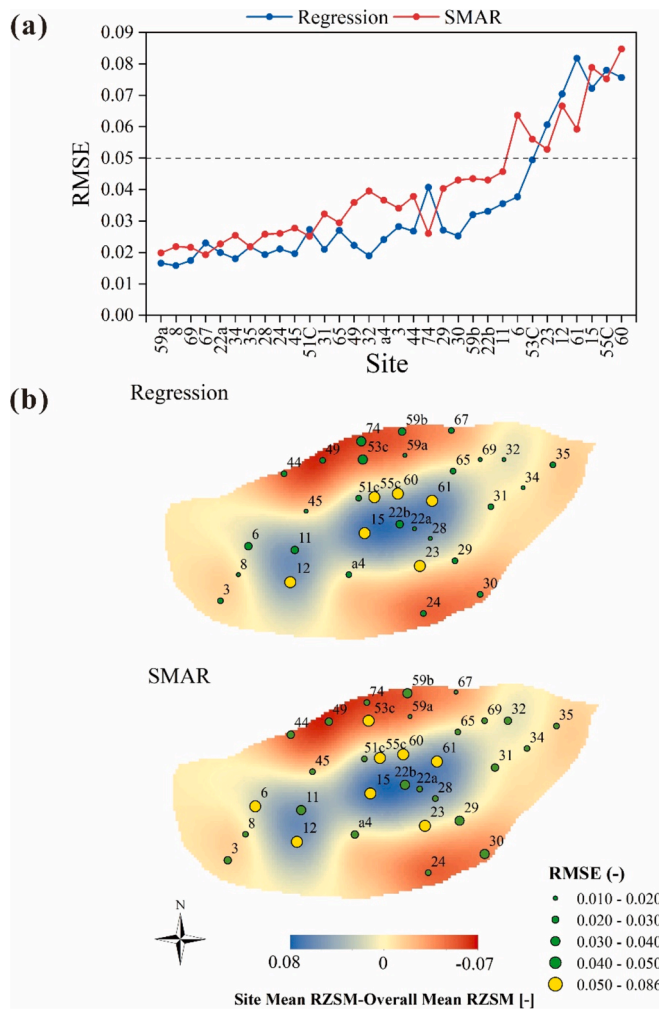


Fig. 6. (a) RMSE(–) of the regression (blue) and SMAR (red) methods from the monitoring stations during the validation period. (b) Maps of the difference between the temporal mean of RZSM(–) observations and the overall mean of these estimations are made using the regression (top) and SMAR (bottom) methods. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

estimations, whereas the regression method still performs well.

Fig. 7b shows a violin plot of the seasonal observed and estimated values. As the soil is drier during the dry–warm season, and, therefore, more sensitive to environmental factors such as rainfall, the moisture values have a wider range during the dry–warm season than during the wet–cold season. The soil is partially sub-saturated during the dry–warm season, whereas it is mostly saturated throughout the wet–cold season. Consequently, there is less variation in the moisture content during the wet–cold season. Overall, there is good agreement between the estimations and observations of RZSM. The comparison of the two estimation approaches shows that the regression method is slightly better than the SMAR method at capturing the RZSM distribution.

Fig. 8a shows the RMSE for each soil unit during the dry–warm and wet–cold seasons. Compared with the dry–warm season, the wet–cold season has a lower RMSE overall. This shows that, during wet periods, there is frequently a stronger link between soil moisture near the surface and in the deep soil. The capacity to simulate RZSM in the dry–warm and wet–cold seasons differs significantly in all units. The Ridgetop and N-slope units generated the most accurate estimations for the two seasons of the six soil-terrain units.

The RMSEs(–) of each site for the two estimation methods during the dry–warm season (top) and the wet–cold season (bottom) are shown in

Fig. 8b. During the dry–warm season, the RMSE(–) between the RZSM estimated by the regression method and the observed RZSM is 0.010 to 0.074, with an average of 0.026. The RMSE(–) of the SMAR method is 0.021 to 0.085, with an average of 0.043. During the wet–cold season, two types of methods show a similar performance. The variations between the sites suggest significant spatial variations in the relationship between the RZSM simulations and observations. Fig. 8b demonstrates that the regression method is a little accurate than the SMAR method at most sites during both the dry–warm and wet–cold seasons.

4.3. Regionalization assessment of two types of methods

The spatial transferability of the two calibrated models was further evaluated to test how well a calibrated regression model at a specific site could be used to make estimations at another site using a regionalization approach (Merz and Blöschl, 2004; Oudin et al., 2008). The mean RMSE(–) was calculated by averaging the estimated results from each donor. Fig. 9 provides a summary of the regionalization results of mean RMSE(–) for the 32 sites in the catchment of two types of methods. Although the regionalization performance of the two types of methods is lower compared to the calibration-based approach, it still demonstrates good performance for ungauged sites. More than half of the sites exhibit a mean RMSE < 0.1, indicating satisfactory assessment accuracy. These results suggest that both methods exhibit good model transferability, underscoring their potential for regionalization. When comparing the two approaches, two types of methods show similar results in several sites, while the regression method provides a slightly more accurate estimation of regionalization, with a lower RMSE at most sites in the catchment.

4.4. Factors influencing RZSM estimation by SSM

The linkage between surface and deep (root zone) soil moisture varies under different environmental conditions. Thus, several factors (e.g., topography, soil properties, and range of soil moisture variation) can impact the estimation of RZSM from SSM. According to how well the site's model performs, study sites are divided into two groups, one with more accurate RZSM estimation (i.e., RMSE < 0.05) and another with less accurate estimation (i.e., RMSE > 0.05). Then, a nonparametric test (Mann–Whitney) is adopted to the statistical difference of 12 selected factors between the two site groups (Table 2). Six variables (the range of SSM variation, range of RZSM variation, average RZSM, PFF, soil depth, and elevation) are significantly different between the two groups, suggesting that these factors likely influence the assessment ability of RZSM in the study catchment.

The most significant factor related to the RZSM estimation accuracy is the range of RZSM variation ($p = 0.003$). That is, a wider range of RZSM fluctuations mean greater variability in the RZSM, which may result in less accurate model simulations. This is mainly reflected in the valley unit which has a thicker soil horizon and uneven soil moisture, having the lowest accuracy among all the sites. Site 59a has relatively stable soil moisture over the three years and generates the most accurate estimations (Fig. 6).

The PFF is also significantly related to the accuracy of RZSM estimation ($p = 0.008$). Previous research has shown that our study sites have preferential flow routes created by biological activity (plant roots or animals; (Liu and Lin, 2015)). The PFF complicates the movement of soil moisture and makes it challenging for the models to accurately simulate these dynamics (Manfreda et al., 2014), so the sites with higher preferential flow network lead to higher RMSE (Table 2). According to the PFF data, depressions on the sunny slopes and the center of the catchment accounted for the majority of the larger PFF values (Liu and Lin, 2015). The spatial distribution of the two types of methods was similar (especially for the SMAR models), indicating that the presence of preferential flow affects the ability of RZSM estimation.

Other significant relationships were found regarding information

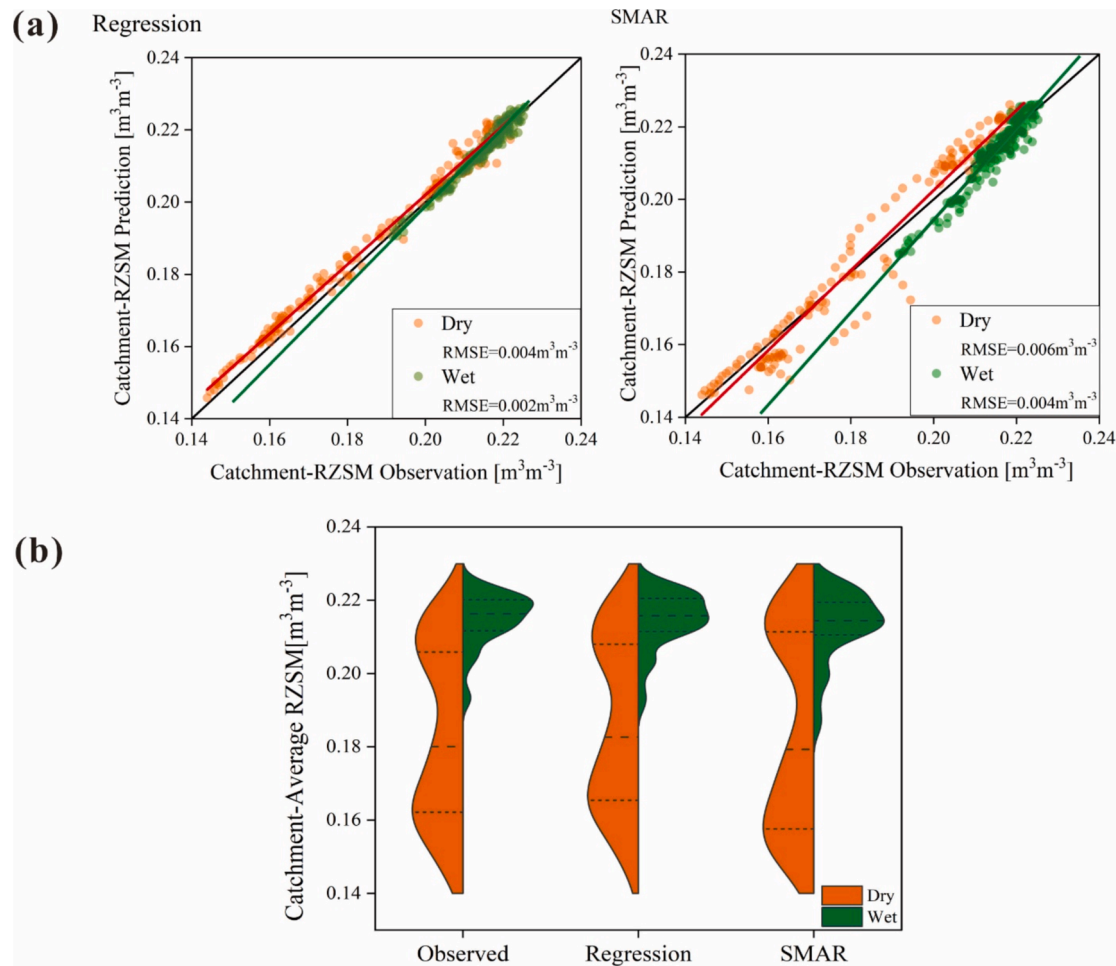


Fig. 7. (a) Scatter plots of estimated and observed RZSM during the dry and wet seasons from the regression (left) and SMAR (right) methods during the validation period. (b) Violin plot of observed and estimated RZSM and statistical table of average RZSM from in situ observations and estimations from both methods during the validation period.

about soil thickness and elevation data. The accuracy of RZSM estimation decreases with increasing soil thickness. This indicates the association between SSM and RZSM is becoming complicated with increasing soil depth. These findings also demonstrate a decline in model estimation precision at sites with lower elevations. This can be explained by the relationship between soil thickness and elevation, which is, the catchment's thicker soils mainly concentrated in the lower-elevation valley bottom area (Baldwin et al., 2019; Grayson et al., 2002; Western et al., 2004). Conversely, SSM content, slope angle, and soil texture have less influence on assessment ability.

5. Discussion

5.1. Comparison between the SMAR model and regression method

The applicability of SMAR to shaded aspects disagreed with some of these earlier findings. For instance, it diverges from the previous observations that SMAR is more effective in arid settings (Guo et al., 2023; Manfreda et al., 2014). Under dry conditions, soil moisture fluxes were vertical (i.e., evapotranspiration and vertical drainage) and localized due to disconnected soil macropores and the amount of solar radiation (Grayson and Western, 1998). This is consistent with the conditions under which the SMAR model is intended to operate, specifically in non-saturated soil moisture conditions and the linear relationship between SSM and RZSM.

However, in humid headwater catchments with complex terrain, the

direction of soil moisture fluxes is more dependent on the microtopography, such as slope, slope curvature, and upslope contributing area (Takagi and Lin, 2011; Takagi and Lin, 2012). This dependence may explain why our study found that the SMAR model is applicable to humid, flat, and shaded aspects. The shaded aspect is flat and has lower spatial variability in microtopography (Fan et al., 2020). Additionally, the soil profiles on the shaded aspect typically contain a higher proportion of rock fragments, which enhances vertical drain (Guo et al., 2019; Sullivan et al., 2019; Fig. 8), aligning with the linear soil loss function assumed by the SMAR model. In contrast, the sunny aspect has extensive root networks and hydrophobic layers (Lin, 2006), creating a developed preferential flow network. This contributed to non-linear soil moisture variations and made it challenging for the SMAR model to accurately represent these dynamics (Manfreda et al., 2014). In conclusion, the homogeneity of soil-terrain contributes to the suitability of the SMAR model for shaded aspects.

Similar to the SMAR model, the regression model also demonstrates superior performance in shaded aspect compared to sunny aspect. However, the regression model exhibits higher simulation accuracy at most sites than the SMAR model. This discrepancy may be attributed to the regression model's reliance on empirically derived relationships constructed from high quality data (Quan, 1988), and the data from the shale hills region is relatively complete and accurate due to long-term maintenance (Fan et al., 2020). Additionally, the relationship between surface soil moisture and root zone soil moisture in the shale hills is influenced by factors with high spatial heterogeneity such as

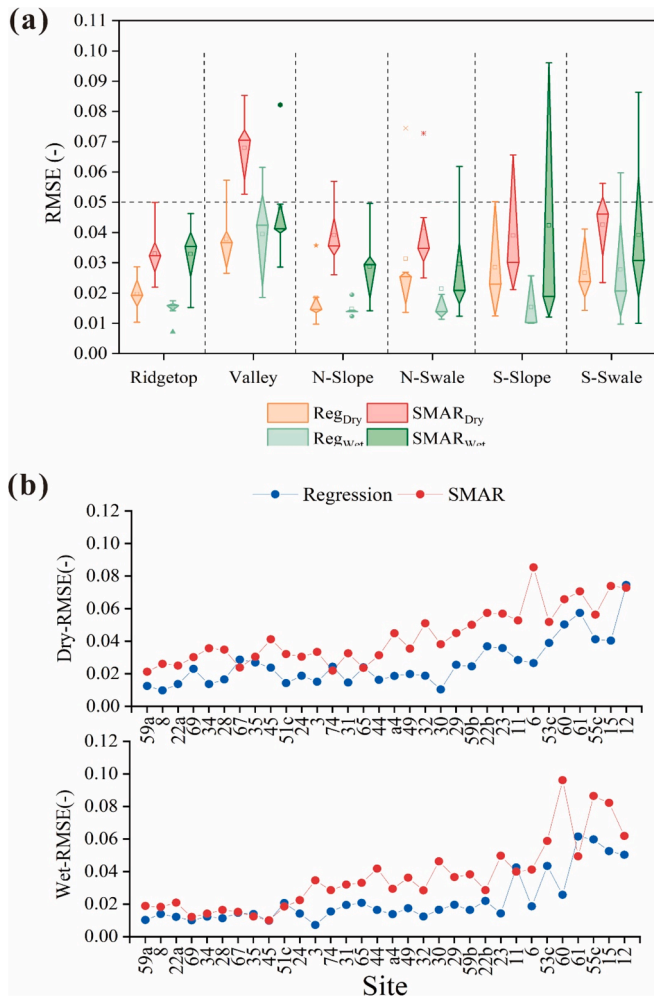


Fig. 8. (a) Box plot of RMSE (–) from the regression and SMAR methods during the dry and wet seasons for the six units during the validation period. (b) RMSE (–) of two types of methods during the dry (top) and wet (bottom) seasons at each site during the validation period.

precipitation, evapotranspiration, and preferential flow (Takagi and Lin, 2011). This relationship is likely to exhibit non-linear variations in soil moisture variation, making it suitable for modeling using quadratic regression models. Han et al. (2011) also indicated that the spatial variability of infiltration and capillary phenomena gives rise to a non-

linear relationship between SSM and RZSM. Consequently, this complexity necessitates the application of filtering techniques and analytical models to accurately characterize and fit this relationship (Zhuang et al., 2020).

Overall, the regression model is more suitable for complex hydro-pedological environments that are supported by long-term and high-quality soil moisture monitoring data. Despite the presence of more intricate hydrological processes, such as non-stationary flows, robust non-linear relationships can be established to estimate RZSM. Conversely, the SMAR model is better suited for flat regions where long-term soil moisture data is lacking. The SMAR model incorporates numerous hydraulic parameters, and a one-time measurement of these parameters, along with simple parameter calibration, can yield satisfactory model performance.

5.2. The relationship between surface soil moisture and root zone soil moisture

All regression models include quadratic terms, indicating that the relationship between surface soil moisture and root zone soil moisture is non-linear (Table S4). Notably, during the dry-warm season, the coefficient a for the valley is greater than zero, suggesting that an increase in SSM leads to a greater increase in RZSM. This phenomenon may be attributed to the valley's lateral saturated hydraulic conductivity being higher than its vertical hydraulic conductivity, resulting in strong water accumulation capacity and a higher frequency of preferential flow during the dry season (Lin, 2006; Liu and Lin, 2015). Consequently, the increase in surface soil moisture enhances lateral hydrological connectivity, providing additional lateral soil moisture to the root zone in the valley. In contrast, during the wet season, the coefficient a for the sunny swale is less than zero, indicating that an increase in SSM results in a reduced increase in RZSM. This could be due to the relatively steep gradient and weaker water accumulation capacity of the sunny swale, which facilitates lateral drainage of surface soil moisture (Fan et al., 2020), thereby diminishing the supply of moisture to the root zone. LIU et al. (2013) employed linear regression models to estimate deep soil moisture; however, they did not account for nonlinear processes such as precipitation and evapotranspiration that influence soil moisture infiltration, leading to considerable uncertainty in their estimates.

In the wet season, the influence of SSM on RZSM is more pronounced. In the regression model, the absolute values of parameters a and b during the wet-cold season are both greater than those in the dry-warm season (Table S4). This indicates that SSM has a more significant impact on RZSM in the wet-cold season. This heightened influence may be attributed to increasing soil water content, which likely leads to stronger percolation driven by increasing unsaturated hydraulic

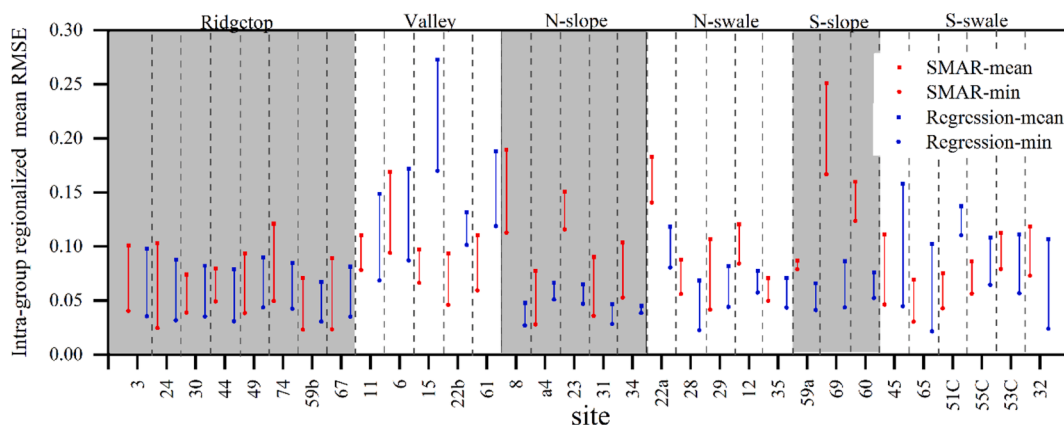


Fig. 9. Performance of spatial transferability of the two calibrated models to make estimations using surface soil moisture measurements at other sites in the catchment using regression (blue) and SMAR (red) methods. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Nonparametric test analysis of different factors influencing model performance.

	Group M (P_{25} , P_{75})		MannWhitney test statistic U	MannWhitney Test statistic Z	p
	Low RMSE(–) (n = 26)	High RMSE(–) (n = 6)			
Range of SSM Variation ($\text{m}^3/\text{m}^{-3}(- -)$)	0.195 (0.2,0.2)	0.261 (0.2,0.3)	25.0	–2.559	0.011*
Average SSM ($\text{m}^3/\text{m}^{-3}(- -)$)	0.218 (0.2,0.2)	0.222 (0.2,0.3)	74.0	–0.193	0.847
Range of RZSM Variation ($\text{m}^3/\text{m}^{-3}(- -)$)	0.125 (0.1,0.1)	0.187 (0.2,0.3)	17.0	–2.945	0.003**
Average RZSM ($\text{m}^3/\text{m}^{-3}(- -)$)	0.200 (0.2,0.2)	0.243 (0.2,0.3)	31.0	–2.269	0.023*
PFF (%)	15.400 (8.6,30.8)	47.375 (37.1,51.2)	21.0	–2.644	0.008**
Slope Curvature (%/100 m)	–0.300 (–3.6,0.7)	–2.920 (–6.1,–0.5)	48.0	–1.449	0.147
Slope Degree (°)	22.000 (14.5,37.1)	19.850 (7.7,28.2)	60.0	–0.869	0.385
Soil Depth (m)	0.500 (0.3,1.5)	1.800 (1.1,2.6)	31.5	–2.255	0.024*
Elevation (m)	283.800 (274.9,291.7)	272.900 (267.0,281.4)	32.0	–2.221	0.026*
Rock (g/cm^3)	44.560 (41.5,58.0)	62.430 (46.3,70.3)	38.0	–1.931	0.053
Sand (g/cm^3)	42.500 (26.8,62.8)	50.000 (22.0,66.5)	77.0	–0.048	0.961
Clay (g/cm^3)	38.235 (33.0,56.3)	59.140 (41.4,76.7)	46.0	–1.545	0.122

* $p < 0.05$ ** $p < 0.01$.

conductivity (Childs and Collis-George, 1950), allowing SSM to transfer more effectively to RZSM. Furthermore, the constant parameter c consistently approaches zero during the wet-cold season. In simpler terms, when the SSM is zero, the value of RZSM also tends to be close to zero, suggesting RZSM is supplied almost entirely by SSM.

Both the SMAR model and regression models exhibit relatively poor estimation performance during the dry-warm season (Fig. 7), which may be attributed to the occurrence of intense rainfall, frequent preferential flow, and significant evapotranspiration during this period. In the shale hills, the dry-warm season is characterized by frequent heavy rainfall, steep terrain, and a dense macropore network created by root channels (Lin, 2006). Additionally, the complexity of root systems and the intensity of evapotranspiration may contribute to strong spatial heterogeneity of soil hydraulic conductivity along the profile (Lin, 2006), thereby reducing the correlation between surface soil moisture and root zone moisture. This perspective is supported by (Ford et al., 2014), who conducted research in the Central Plains of the United States. However, LIU et al. (2013) found that the magnitude of soil water storage itself does not significantly affect estimation performance. If the impact is attributed to differences in magnitude, it reflects the influence of ecosystem factors rather than the magnitude itself.

5.3. Outlook

In this study, we used data covering two years to establish a functional relationship between SSM and RZSM and data from an additional single year for validation. At study sites without such a long period of data coverage, determining whether the regression method still outperforms the SMAR model would be more difficult. Recent studies have adopted machine learning algorithms to develop the estimation model for RZSM (e.g., (Carranza et al., 2021; Kornelsen and Coulibaly, 2014)). Considering the strong computing power and learning advantages of machine learning, machine learning can also be incorporated into the comparison of different spatial and temporal scales of the RZSM estimation models in future studies, to screen more reasonable estimation models for soil moisture prediction. Future research is suggested to explore the integration of physical process-based methods with machine learning methods. The constraints of physical mechanism-based methods in relying on input data coverage and quality, selection of key model parameters, and the determination of loss functions, can be partially be addressed by incorporating machine learning techniques (Read et al., 2019).

This study explored the potential of RZSM estimation using the SMAR and regression methods at gauged and ungauged sites. However, RZSM varies widely with sites/regions in complex environments and is influenced by catchment properties such as vegetation, land use, topography, and soil properties. So, it still needs further studies to explore how sites can be better grouped and which site properties are

important for grouping as moving toward RZSM estimation in ungauged regions. Specially, when preferential flow dominates the infiltration process during the dry-warm season, the accuracy of both methods declined. Thus, it will be necessary to develop further hydrological models that can describe preferential flow and estimate the corresponding soil moisture distribution. Alternatively, factors controlling preferential flow mechanisms could be identified to establish a multi-variable model to estimate RZSM from SSM during rainfall events.

6. Conclusions

This study compared the SMAR model and regression method for estimating RZSM from SSM in areas with complex topography. We examined the applicability of two methods at different spatial scales and during different seasons across 32 monitoring sites in a forested catchment over three years. The results show that both methods can generate accurate estimations of RZSM with calibration at gauged sites and have a good potential for estimations at ungauged regions. The estimation ability of the methods varies spatially and is better in the drier areas than in the wetter areas of the catchment. The performance of both approaches also varies temporally, and the RZSM is better estimated during the wet-cold season than during the dry-warm season. Our comparison of the two approaches demonstrates that the two models show comparable performance in the study area (with NSE values of 0.986 and 0.951 during the validation period, respectively). It indicates that it may be advantageous to apply regression method in regions with non-linear hydrogeological behaviors and long-term datasets, whereas the SMAR model may be more suitable for relatively flat areas with less spatial variability in microtopography. We also reveal that model performance is significantly impacted by variables such as soil moisture range, terrain and preferential flow. The results of this study will assist with model selection for RZSM estimation in mountainous catchments and will also improve our understanding of soil moisture dynamics. Further development of SMAR model that can describe preferential flow and estimate the corresponding soil moisture distribution will improve RZSM estimations in future studies. We also suggest to explore the integration of physical process-based methods with machine learning techniques, which may facilitate better methods for future RZSM estimation work.

CRedit authorship contribution statement

Yongliang Qi: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Bihang Fan:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yaling Zhang:** Writing – review & editing, Data curation. **Yanjia Jiang:** Investigation, Formal analysis. **Yuanyuan Huang:** Validation,

Resources. **Elizabeth W. Boyer**: Writing – review & editing, Validation. **Carlos R. Mello**: Writing – review & editing, Validation. **Li Guo**: Writing – review & editing, Validation, Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization. **Hongxia Li**: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2024.117144>.

Data availability

Data will be made available on request.

References

- Andrade, N.P., Viola, M.R., Beskow, S., Caldeira, T.L., Guo, L., Mello, C.R., 2020. Assessment of spatial and temporal soil water storage using a distributed hydrological model. *Water Resources Management* 34, 5031–5046.
- Baldwin, D., Manfreda, S., Keller, K., Smithwick, E., 2017. Predicting root zone soil moisture with soil properties and satellite near-surface moisture data across the conterminous United States. *Journal of Hydrology* 546, 393–404.
- Baldwin, D., Manfreda, S., Lin, H., Smithwick, E.A.H., 2019. Estimating Root Zone Soil Moisture Across the Eastern United States with Passive Microwave Satellite Data and a Simple Hydrologic Model. *Remote Sensing* 11 (17).
- Bogena, H.R., Huisman, J.A., Oberdörster, C., Vereecken, H., 2007. Evaluation of a low-cost soil water content sensor for wireless network applications. *Journal of Hydrology* 344 (1–2), 32–42.
- Carranza, C., Nolet, C., Peziz, M., van der Ploeg, M., 2021. Root zone soil moisture estimation with Random Forest. *Journal of Hydrology* 593, 125840.
- Childs, E.C., Collis-George, N., 1950. The permeability of porous materials. *Proceedings of the royal society of London. series a. mathematical and physical sciences* 201 (1066), 392–405.
- Das, N.N., Mohanty, B.P., 2006. Root zone soil moisture assessment using remote sensing and vadose zone modeling. *Vadose Zone Journal* 5 (1), 296–307.
- Dobriyal, P., Qureshi, A., Badola, R., Hussain, S.A., 2012. A review of the methods available for estimating soil moisture and its implications for water resource management. *Journal of Hydrology* 458, 110–117.
- Dong, S.-H., 2008. Genetic algorithm based parameter estimation of Nash model. *Water Resources Management* 22, 525–533.
- Elshorbagy, A., Parasuraman, K., 2008. On the relevance of using artificial neural networks for estimating soil moisture content. *Journal of Hydrology* 362 (1–2), 1–18.
- Fan, B., Tao, W., Qin, G., Hopkins, I., Zhang, Y., Wang, Q., Lin, H., Guo, L., 2020. Soil micro-climate variation in relation to slope aspect, position, and curvature in a forested catchment. *Agricultural and Forest Meteorology* 290, 107999.
- Faridani, F., Farid, A., Ansari, H., Manfreda, S., 2017. Estimation of the root-zone soil moisture using passive microwave remote sensing and SMAR Model. *Journal of Irrigation and Drainage Engineering* 143 (1), 04016070.
- Farokhi, M., Faridani, F., Lasaponara, R., Ansari, H., Faridhosseini, A., 2021. Enhanced Estimation of Root Zone Soil Moisture at 1 km Resolution Using SMAR Model and MODIS-Based Downscaled AMSR2 Soil Moisture Data. *Sensors* 21 (15), 5211.
- Ford, T., Harris, E., Quiring, S., 2014. Estimating root zone soil moisture using near-surface observations from SMOS. *Hydrology and Earth System Sciences* 18 (1), 139–154.
- Grayson, R.B., Blöschl, G., Western, A.W., McMahon, T.A., 2002. Advances in the use of observed spatial patterns of catchment hydrological response. *Advances in Water Resources* 25 (8–12), 1313–1334.
- Grayson, R.B., Western, A.W., 1998. Towards areal estimation of soil water content from point measurements: time and space stability of mean response. *Journal of Hydrology* 207 (1–2), 68–82.
- Guo, X., Fang, X., Zhu, Q., Jiang, S., Tian, J., Tian, Q., Jin, J., 2023. Estimation of Root-Zone Soil Moisture in Semi-Arid Areas Based on Remotely Sensed Data. *Remote Sensing* 15 (8), 2003.
- Guo, L., Lin, H., Fan, B., Nyquist, J., Toran, L., Mount, G.J., 2019. Preferential flow through shallow fractured bedrock and a 3D fill-and-spill model of hillslope subsurface hydrology. *Journal of Hydrology* 576, 430–442.
- Han, E., Merwade, V., Heathman, G.C., 2011. Application of data assimilation with the Root Zone Water Quality Model for soil moisture profile estimation in the upper Cedar Creek. *Indiana. Hydrological Processes* 26 (11), 1707–1719.
- Hasenmueller, E.A., Gu, X., Weitzman, J.N., Adams, T.S., Stinchcomb, G.E., Eissenstat, D. M., Drohan, P.J., Brantley, S.L., Kaye, J.P., 2017. Weathering of rock to regolith: The activity of deep roots in bedrock fractures. *Geoderma* 300, 11–31.
- Kornelsen, K.C., Coulbaly, P., 2014. Root-zone soil moisture estimation using data-driven methods. *Water Resources Research* 50 (4), 2946–2962.
- Kurc, S.A., Small, E.E., 2007. Soil moisture variations and ecosystem-scale fluxes of water and carbon in semiarid grassland and shrubland. *Water Resources Research* 43 (6).
- Li, Z.-L., Leng, P., Zhou, C., Chen, K.-S., Zhou, F.-C., Shang, G.-F., 2021. Soil moisture retrieval from remote sensing measurements: Current knowledge and directions for the future. *Earth-Science Reviews* 218, 103673.
- Li, H., Zhang, Y., 2017. Regionalising rainfall-runoff modelling for predicting daily runoff: Comparing gridded spatial proximity and gridded integrated similarity approaches against their lumped counterparts. *Journal of Hydrology* 550, 279–293.
- Li, H., Zhang, Y., Chiew, F.H., Xu, S., 2009. Predicting runoff in ungauged catchments by using Xinanjiang model with MODIS leaf area index. *Journal of Hydrology* 370 (1–4), 155–162.
- Lin, H., 2006. Temporal stability of soil moisture spatial pattern and subsurface preferential flow pathways in the Shale Hills Catchment. *Vadose Zone Journal* 5 (1), 317–340.
- Liu, Y., Li, T., Zhang, W., Lv, A., 2023. Regionalization of Root Zone Moisture Estimations from Downscaled Surface Moisture and Environmental Data with the Soil Moisture Analytical Relationship Model. *Water* 15 (23), 4133.
- Liu, H., Lin, H., 2015. Frequency and control of subsurface preferential flow: From pedon to catchment scales. *Soil Science Society of America Journal* 79 (2), 362–377.
- Liu, S.-X., Xing, B., Yuan, G.-F., Mo, X.-G., Lin, Z.-H., 2013. Relationship analysis between soil moisture in root zone and top-most layer in China. *Chinese Journal of Plant Ecology* 37 (1), 1.
- Ma, L., Chabaux, F., West, N., Kirby, E., Jin, L., Brantley, S., 2013. Regolith production and transport in the Susquehanna Shale Hills Critical Zone Observatory, part 1: insights from U-series isotopes. *Journal of Geophysical Research: Earth Surface* 118 (2), 722–740.
- Magagi, R., Kerr, Y., 2001. Estimating surface soil moisture and soil roughness over semiarid areas from the use of the copolarization ratio. *Remote Sensing of Environment* 75 (3), 432–445.
- Mahmood, R., Hubbard, K.G., 2007. Relationship between soil moisture of near surface and multiple depths of the root zone under heterogeneous land uses and varying hydroclimatic conditions. *Hydrological Processes: an International Journal* 21 (25), 3449–3462.
- Mahmood, R., Littell, A., Hubbard, K.G., You, J., 2012. Observed data-based assessment of relationships among soil moisture at various depths, precipitation, and temperature. *Applied Geography* 34, 255–264.
- Manfreda, S., Brocca, L., Moramarco, T., Melone, F., Sheffield, J., 2014. A physically based approach for the estimation of root-zone soil moisture from surface measurements. *Hydrology and Earth System Sciences* 18 (3), 1199–1212.
- Merz, R., Blöschl, G., 2004. Regionalisation of catchment model parameters. *Journal of Hydrology* 287 (1–4), 95–123.
- Oke, U., 2009. Statistical analysis of the application of Wilcoxon and Mann-Whitney U test in medical research studies. *Biotechnology and Molecular Biology Reviews* 4 (6), 128–131.
- Oudin, L., Andréassian, V., Perrin, C., Michel, C., Le Moine, N., 2008. Spatial proximity, physical similarity, regression and ungauged catchments: A comparison of regionalization approaches based on 913 French catchments. *Water Resources Research* 44 (3).
- Pan, X., Kornelsen, K.C., Coulbaly, P., 2017. Estimating root zone soil moisture at continental scale using neural networks. *JAWRA Journal of the American Water Resources Association* 53 (1), 220–237.
- Parajka, J., Merz, R., Blöschl, G., 2005. A comparison of regionalisation methods for catchment model parameters. *Hydrology and Earth System Sciences* 9 (3), 157–171.
- Pauwels, V.R., Hoeben, R., Verhoest, N.E., De Troch, F.P., 2001. The importance of the spatial patterns of remotely sensed soil moisture in the improvement of discharge predictions for small-scale basins through data assimilation. *Journal of Hydrology* 251 (1–2), 88–102.
- Peziz, M., Augustijn, D.C., Hendriks, D.M., Hulscher, S.J., 2020. Applying transfer function-noise modelling to characterize soil moisture dynamics: a data-driven approach using remote sensing data. *Environmental Modelling & Software* 131, 104756.
- Primka IV, E.J., Adams, T.S., Buck, A., Eissenstat, D.M., 2021. Topographical shifts in fine root lifespan in a mixed, mesic temperate forest. *Plos One* 16 (7), e0254672.
- Primka, E.J., Adams, T.S., Buck, A.S., Eissenstat, D.M., 2022. Shifts in root dynamics along a hillslope in a mixed, mesic temperate forest. *Plant and Soil* 1–17.
- Quan, N.T., 1988. The prediction sum of squares as a general measure for regression diagnostics. *Journal of Business & Economic Statistics* 6 (4), 501–504.
- Read, J.S., Jia, X., Willard, J., Appling, A.P., Zwart, J.A., Oliver, S.K., Karpatne, A., Hansen, G.J., Hanson, P.C., Watkins, W., 2019. Process-guided deep learning

- predictions of lake water temperature. *Water Resources Research* 55 (11), 9173–9190.
- Rigden, A., Mueller, N., Holbrook, N., Pillai, N., Huybers, P., 2020. Combined influence of soil moisture and atmospheric evaporative demand is important for accurately predicting US maize yields. *Nature Food* 1 (2), 127–133.
- Schmugge, T.J., Kustas, W.P., Ritchie, J.C., Jackson, T.J., Rango, A., 2002. Remote sensing in hydrology. *Advances in Water Resources* 25 (8–12), 1367–1385.
- Smith, L.A., Eissenstat, D.M., Kaye, M.W., 2017. Variability in aboveground carbon driven by slope aspect and curvature in an eastern deciduous forest, USA. *Canadian Journal of Forest Research* 47 (2), 149–158.
- Srivastava, P.K., Han, D., Ramirez, M.A.R., Islam, T., 2013. Appraisal of SMOS soil moisture at a catchment scale in a temperate maritime climate. *Journal of Hydrology* 498, 292–304.
- Sullivan, P., Godd  ris, Y., Shi, Y., Gu, X., Schott, J., Hasenmueller, E., Kaye, J., Duffy, C., Jin, L., Brantley, S., 2019. Exploring the effect of aspect to inform future earthcasts of climate-driven changes in weathering of shale. *Journal of Geophysical Research: Earth Surface* 124 (4), 974–993.
- Takagi, K., Lin, H., 2011. Temporal dynamics of soil moisture spatial variability in the Shale Hills Critical Zone Observatory. *Vadose Zone Journal* 10 (3), 832–842.
- Takagi, K., Lin, H., 2012. Changing controls of soil moisture spatial organization in the Shale Hills Catchment. *Geoderma* 173, 289–302.
- Vereecken, H., Huisman, J., Boga, H., Vanderborght, J., Vrugt, J., Hopmans, J., 2008. On the value of soil moisture measurements in vadose zone hydrology: A review. *Water Resources Research* 44 (4).
- Wang, X., Xie, H., Guan, H., Zhou, X., 2007. Different responses of MODIS-derived NDVI to root-zone soil moisture in semi-arid and humid regions. *Journal of Hydrology* 340 (1–2), 12–24.
- Western, A.W., Grayson, R.B., Bl  schl, G., 2002. Scaling of soil moisture: A hydrologic perspective. *Annual Review of Earth and Planetary Sciences* 30 (1), 149–180.
- Western, A.W., Zhou, S.-L., Grayson, R.B., McMahon, T.A., Bl  schl, G., Wilson, D.J., 2004. Spatial correlation of soil moisture in small catchments and its relationship to dominant spatial hydrological processes. *Journal of Hydrology* 286 (1–4), 113–134.
- Wilson, D.J., Western, A.W., Grayson, R.B., Berg, A.A., Lear, M.S., Rodell, M., Famiglietti, J.S., Woods, R.A., McMahon, T.A., 2003. Spatial distribution of soil moisture over 6 and 30 cm depth, Mahurangi river catchment, New Zealand. *Journal of Hydrology* 276 (1–4), 254–274.
- Zhuang, R., Zeng, Y., Manfreda, S., Su, Z., 2020. Quantifying Long-Term Land Surface and Root Zone Soil Moisture over Tibetan Plateau. *Remote Sensing* 12 (3).